

Total No. of Questions : 4]

SEAT No. :

PC413

[Total No. of Pages : 1

[6359]-533

S.E. (AI & ML) (Insem)

DATASTRUCTURES & ALGORITHMS

(2019 Pattern) (Semester - III) (218542)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Answer Q1 or Q2, Q3 or Q4.
- 2) Figures to the right indicate full marks.
- 3) Assume suitable data, if necessary.

- Q1)** a) Write sudo code for inserting an element in doubly linked list. [6]
b) Discuss various asymptotic notations used to analyze algorithms. [6]
c) Enlist differences between Data & Data object. [3]

OR

- Q2)** a) Explain with the help of examples how row-major & column-major address calculations are performed for multi-dimensional arrays. [8]
b) Write sudo code delete a specified element from a singly circular list. [7]

- Q3)** a) Discuss various sorting-algorithm characteristics. [3]
b) Establish comparative analysis of merge sort & Quick sort algorithms. [6]
c) Write sudo code & dry-run for binary search algorithm. [6]

OR

- Q4)** a) Discuss the importance of sort-stability with respect to sorting algorithms. [3]
b) Discuss the time complexity of following sorting algorithm. [6]
i) Shell sort.
ii) Merge sort.
iii) Quick sort.
c) Write sudo code & dryrun for fibonacci search. [6]

